



```
In [54... #import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [55... events=pd.read_csv("DS_events.csv")
```

```
C:\Users\Deeksha\AppData\Local\Temp\ipykernel_26676\2863476826.py:1: DtypeWarning: Columns (10,29,32,33,38,39) have mixed types. Specify dtype option on import or set low_memory=False.
events=pd.read_csv("DS_events.csv")
```

```
In [56... events.shape
```

```
Out[56]: (66568, 41)
```

```
In [57... events.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 66568 entries, 0 to 66567
Data columns (total 41 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Column1                                   66568 non-null  int64
1   clock                                    66568 non-null  object
2   defender_id                             14055 non-null  float64
3   defenders                                0 non-null      float64
4   defending_all_out_points                  19799 non-null  float64
5   defending_bonus_points                   19799 non-null  float64
6   defending_capture_points                  19799 non-null  float64
7   defending_points                          58315 non-null  float64
8   defending_team_id                        58315 non-null  float64
9   defending_technical_points               19799 non-null  float64
10  do_or_die                               20297 non-null  object
11  event                                    66568 non-null  object
12  event_id                                66568 non-null  int64
13  event_no                                66568 non-null  int64
14  event_text                              66568 non-null  object
15  match_id                                66568 non-null  int64
16  player_id                               5441 non-null   float64
17  raid_all_out_points                     19799 non-null  float64
18  raid_bonus_points                       19799 non-null  float64
19  raid_points                             58315 non-null  float64
20  raid_technical_points                   19799 non-null  float64
21  raid_touch_points                       19799 non-null  float64
22  raider_bonus                            498 non-null    float64
23  raider_id                               58315 non-null  float64
24  raiding_team_id                         58315 non-null  float64
25  reason                                  922 non-null    object
26  reason_id                               0 non-null      float64
27  review.outcome                          0 non-null      float64
28  review.outcome.id                       443 non-null    float64
29  review.outcome.outcome_value            443 non-null    object
30  review.reason                            0 non-null      float64
31  review.reason.id                        443 non-null    float64
32  review.reason.reason_value              443 non-null    object
33  review.review_taken                     19799 non-null  object
34  review.team_id                          19799 non-null  float64
35  score                                   59233 non-null  object
36  status_id                               45112 non-null  float64
37  substituted_by                           4733 non-null   float64
38  super_raid                             19799 non-null  object
39  super_tackle                            19799 non-null  object
40  team_id                                 8250 non-null   float64
dtypes: float64(26), int64(4), object(11)
memory usage: 20.8+ MB

```

```
In [58... events.isnull().sum()
```

```

Out[58]: Column1          0
        clock            0
        defender_id      52513
        defenders        66568
        defending_all_out_points 46769
        defending_bonus_points 46769
        defending_capture_points 46769
        defending_points    8253
        defending_team_id   8253
        defending_technical_points 46769
        do_or_die         46271
        event             0
        event_id          0
        event_no          0
        event_text        0
        match_id          0
        player_id        61127
        raid_all_out_points 46769
        raid_bonus_points 46769
        raid_points       8253
        raid_technical_points 46769
        raid_touch_points 46769
        raider_bonus      66070
        raider_id         8253
        raiding_team_id   8253
        reason            65646
        reason_id         66568
        review.outcome    66568
        review.outcome.id 66125
        review.outcome.outcome_value 66125
        review.reason     66568
        review.reason.id  66125
        review.reason.reason_value 66125
        review.review_taken 46769
        review.team_id    46769
        score             7335
        status_id         21456
        substituted_by     61835
        super_raid        46769
        super_tackle      46769
        team_id           58318
        dtype: int64

```

```

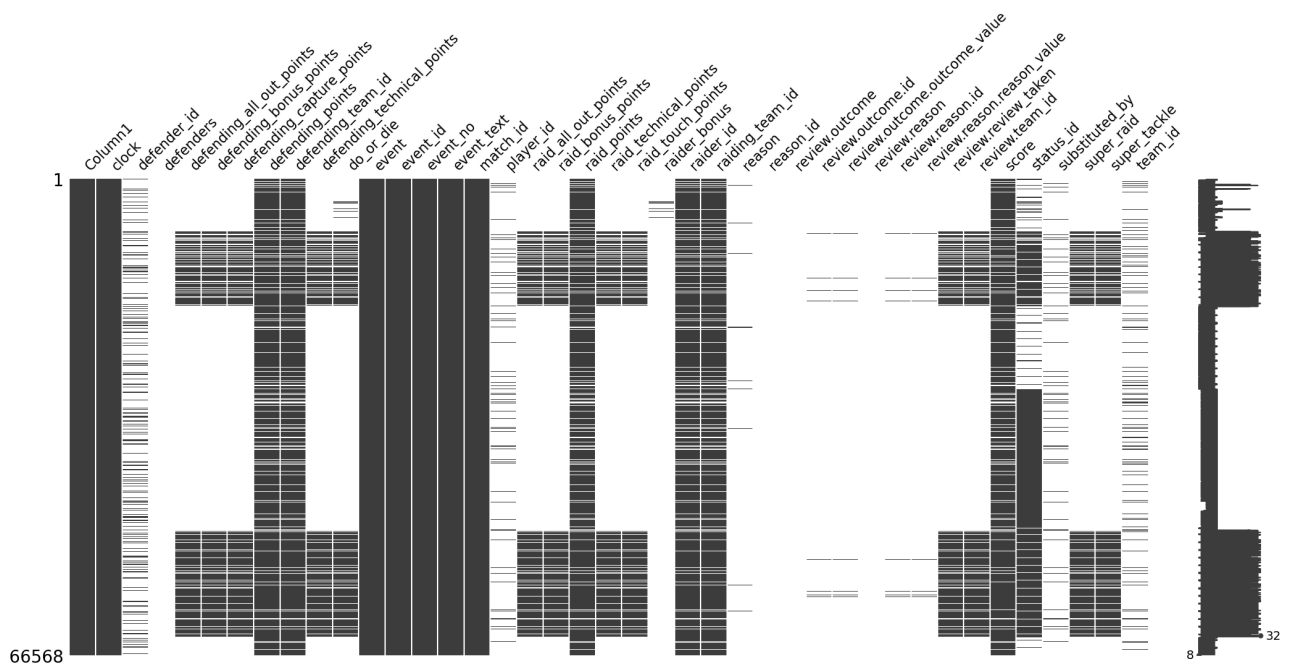
In [59... !pip install missingno

```

Requirement already satisfied: missingno in c:\users\deeksha\anaconda3\lib\site-packages (0.5.2)
Requirement already satisfied: numpy in c:\users\deeksha\anaconda3\lib\site-packages (from missingno) (2.3.5)
Requirement already satisfied: matplotlib in c:\users\deeksha\anaconda3\lib\site-packages (from missingno) (3.10.6)
Requirement already satisfied: scipy in c:\users\deeksha\anaconda3\lib\site-packages (from missingno) (1.16.3)
Requirement already satisfied: seaborn in c:\users\deeksha\anaconda3\lib\site-packages (from missingno) (0.13.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\deeksha\anaconda3\lib\site-packages (from matplotlib->missingno) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\deeksha\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->missingno) (1.17.0)
Requirement already satisfied: pandas>=1.2 in c:\users\deeksha\anaconda3\lib\site-packages (from seaborn->missingno) (2.3.3)
Requirement already satisfied: pytz>=2020.1 in c:\users\deeksha\anaconda3\lib\site-packages (from pandas>=1.2->seaborn->missingno) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\deeksha\anaconda3\lib\site-packages (from pandas>=1.2->seaborn->missingno) (2025.2)

```
In [60... #we can visualize the null values  
import missingno as msno
```

```
In [61... msno.matrix(events)  
plt.show()
```



```
In [62...] events.describe() #numerical columns
```

```
Out[62]:
```

	Column1	defender_id	defenders	defending_all_out_points	defending_bonus_po
count	66568.000000	14055.000000	0.0	19799.000000	19799.000
mean	48.710326	386.817645	NaN	0.028486	0.021
std	28.798869	557.652491	NaN	0.236989	0.143
min	0.000000	4.000000	NaN	0.000000	0.000
25%	24.000000	96.000000	NaN	0.000000	0.000
50%	48.000000	179.000000	NaN	0.000000	0.000
75%	73.000000	401.000000	NaN	0.000000	0.000
max	122.000000	2672.000000	NaN	2.000000	1.000

8 rows × 30 columns

```
In [63...] events.dtypes
```

```

Out[63]: Column1                                int64
         clock                                object
         defender_id                        float64
         defenders                          float64
         defending_all_out_points            float64
         defending_bonus_points             float64
         defending_capture_points           float64
         defending_points                   float64
         defending_team_id                  float64
         defending_technical_points         float64
         do_or_die                         object
         event                            object
         event_id                         int64
         event_no                         int64
         event_text                       object
         match_id                        int64
         player_id                       float64
         raid_all_out_points              float64
         raid_bonus_points                float64
         raid_points                      float64
         raid_technical_points            float64
         raid_touch_points                float64
         raider_bonus                     float64
         raider_id                       float64
         raiding_team_id                  float64
         reason                          object
         reason_id                       float64
         review.outcome                   float64
         review.outcome.id                float64
         review.outcome.outcome_value     object
         review.reason                    float64
         review.reason.id                 float64
         review.reason.reason_value       object
         review.review_taken              object
         review.team_id                   float64
         score                           object
         status_id                       float64
         substituted_by                   float64
         super_raid                       object
         super_tackle                     object
         team_id                          float64
         dtype: object

```

```

In [64... #nunique()
events.nunique()

```

```

Out[64]: Column1      123
         clock      3212
         defender_id  536
         defenders      0
         defending_all_out_points  2
         defending_bonus_points  2
         defending_capture_points  2
         defending_points  4
         defending_team_id  45
         defending_technical_points  3
         do_or_die      2
         event      10
         event_id      10
         event_no      123
         event_text    5413
         match_id      684
         player_id     610
         raid_all_out_points  2
         raid_bonus_points  2
         raid_points     12
         raid_technical_points  5
         raid_touch_points  6
         raider_bonus     2
         raider_id     631
         raiding_team_id  45
         reason      20
         reason_id      0
         review.outcome  0
         review.outcome.id  2
         review.outcome.outcome_value  2
         review.reason  0
         review.reason.id  8
         review.reason.reason_value  8
         review.review_taken  2
         review.team_id  20
         score      2438
         status_id      9
         substituted_by  642
         super_raid      2
         super_tackle     2
         team_id      45
         dtype: int64

```

```

In [65... events["super_tackle"].value_counts()

```

```

Out[65]: super_tackle
         False    19350
         True      449
         Name: count, dtype: int64

```

```

In [66... events.describe(include="all")

```

Out[66]:

	Column1	clock	defender_id	defenders	defending_all_out_points	defending_b
count	66568.000000	66568	14055.000000	0.0	19799.000000	1
unique	NaN	3212	NaN	NaN	NaN	
top	NaN	19:58	NaN	NaN	NaN	
freq	NaN	303	NaN	NaN	NaN	
mean	48.710326	NaN	386.817645	NaN	0.028486	
std	28.798869	NaN	557.652491	NaN	0.236989	
min	0.000000	NaN	4.000000	NaN	0.000000	
25%	24.000000	NaN	96.000000	NaN	0.000000	
50%	48.000000	NaN	179.000000	NaN	0.000000	
75%	73.000000	NaN	401.000000	NaN	0.000000	
max	122.000000	NaN	2672.000000	NaN	2.000000	

11 rows × 41 columns

```
In [67... #Data cleaning  
#data type conversion  
events["super_tackle"].value_counts()
```

```
Out[67]: super_tackle  
False    19350  
True       449  
Name: count, dtype: int64
```

```
In [68... events["super_tackle"]=events['super_tackle'].astype(int)
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[68], line 1
----> 1 events["super_tackle"] = events[          ].astype(int)

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:6665, in NDFrame
e.astype(self, dtype, copy, errors)
    6659         results = [
    6660             ser.astype(dtype, copy=copy, errors=errors) for _, ser in s
elf.items()
    6661         ]
    6663     else:
    6664         # else, only a single dtype is given
-> 6665         new_data = self._mgr.astype(dtype=dtype, copy=copy, errors=erro
rs)
    6666         res = self._constructor_from_mgr(new_data, axes=new_data.axes)
    6667         return res.__finalize__(self, method="astype")

File ~\anaconda3\Lib\site-packages\pandas\core\internals\managers.py:449, i
n BaseBlockManager.astype(self, dtype, copy, errors)
    446     elif using_copy_on_write():
    447         copy = False
-> 449     return self.apply(
    450         ,
    451         dtype=dtype,
    452         copy=copy,
    453         errors=errors,
    454         using_cow=using_copy_on_write(),
    455     )

File ~\anaconda3\Lib\site-packages\pandas\core\internals\managers.py:363, i
n BaseBlockManager.apply(self, f, align_keys, **kwargs)
    361         applied = b.apply(f, **kwargs)
    362     else:
-> 363         applied = getattr(b, f)(**kwargs)
    364         result_blocks = extend_blocks(applied, result_blocks)
    366     out = type(self).from_blocks(result_blocks, self.axes)

File ~\anaconda3\Lib\site-packages\pandas\core\internals\blocks.py:784, in
Block.astype(self, dtype, copy, errors, using_cow, squeeze)
    781         raise ValueError("Can not squeeze with more than one column
n.")
    782         values = values[0, :] # type: ignore[call-overload]
-> 784     new_values = astype_array_safe(values, dtype, copy=copy, errors=err
ors)
    786     new_values = maybe_coerce_values(new_values)
    788     refs = None

File ~\anaconda3\Lib\site-packages\pandas\core\dtypes\astype.py:237, in ast
ype_array_safe(values, dtype, copy, errors)
    234     dtype = dtype.numpy_dtype
    236     try:
-> 237         new_values = astype_array(values, dtype, copy=copy)
    238     except (ValueError, TypeError):

```

```

239     # e.g. _astype_nansafe can fail on object-dtype of strings
240     # trying to convert to float
241     if errors == "ignore":

```

File ~\anaconda3\Lib\site-packages\pandas\core\dtypes\astype.py:182, in `astype_array(values, dtype, copy)`

```

179     values = values.astype(dtype, copy=copy)
181 else:
--> 182     values = _astype_nansafe(values, dtype, copy=copy)
184 # in pandas we don't store numpy str dtypes, so convert to object
185 if isinstance(dtype, np.dtype) and issubclass(values.dtype.type, str):

```

File ~\anaconda3\Lib\site-packages\pandas\core\dtypes\astype.py:133, in `_astype_nansafe(arr, dtype, copy, skipna)`

```

129     raise ValueError(msg)
131 if copy or arr.dtype == object or dtype == object:
132     # Explicit copy, or required since NumPy can't view from / to object.
--> 133     return arr.astype(dtype, copy=True)
135 return arr.astype(dtype, copy=copy)

```

ValueError: cannot convert float NaN to integer

```
In [69... events["super_tackle"]=pd.to_numeric(events["super_tackle"],errors="coerc
```

```
In [70... events["super_raid"]=pd.to_numeric(events["super_raid"])
```

```
In [71... #data type conversion
#categorical data
#select all the object columns
categorical_columns=events.select_dtypes(include=["object"]).columns

```

```
In [72... categorical_columns
```

```
Out[72]: Index(['clock', 'do_or_die', 'event', 'event_text', 'reason',
              'review.outcome.outcome_value', 'review.reason.reason_value',
              'review.review_taken', 'score'],
              dtype='object')
```

```
In [73... #convert into category
for c in categorical_columns:
    events[c]=events[c].astype("category")

```

```
In [74... events.dtypes
```

```

Out[74]: Column1                                int64
clock                                           category
defender_id                                    float64
defenders                                       float64
defending_all_out_points                       float64
defending_bonus_points                        float64
defending_capture_points                     float64
defending_points                             float64
defending_team_id                             float64
defending_technical_points                   float64
do_or_die                                      category
event                                           category
event_id                                       int64
event_no                                       int64
event_text                                    category
match_id                                       int64
player_id                                     float64
raid_all_out_points                          float64
raid_bonus_points                           float64
raid_points                                  float64
raid_technical_points                       float64
raid_touch_points                           float64
raider_bonus                                 float64
raider_id                                    float64
raiding_team_id                             float64
reason                                         category
reason_id                                    float64
review.outcome                              float64
review.outcome.id                           float64
review.outcome.outcome_value                 category
review.reason                               float64
review.reason.id                            float64
review.reason.reason_value                   category
review.review_taken                          category
review.team_id                              float64
score                                         category
status_id                                   float64
substituted_by                              float64
super_raid                                  float64
super_tackle                                float64
team_id                                      float64
dtype: object

```

```

In [75... #identify duplicates
duplicates=events.duplicated()

```

```

In [76... duplicates

```

```
Out[76]: 0      False
          1      False
          2      False
          3      False
          4      False
          ...
          66563   False
          66564   False
          66565   False
          66566   False
          66567   False
          Length: 66568, dtype: bool
```

```
In [77... duplicates.sum()
```

```
Out[77]: np.int64(0)
```

```
In [78... #drop_duplicates()
          # events=events.drop_duplicates()
```

```
In [79... #dropna()
          events.dropna(axis=1,how="all",inplace=True)
```

```
In [80... events.columns
```

```
Out[80]: Index(['Column1', 'clock', 'defender_id', 'defending_all_out_points',
                'defending_bonus_points', 'defending_capture_points',
                'defending_points', 'defending_team_id', 'defending_technical_po
ints',
                'do_or_die', 'event', 'event_id', 'event_no', 'event_text', 'mat
ch_id',
                'player_id', 'raid_all_out_points', 'raid_bonus_points', 'raid_p
oints',
                'raid_technical_points', 'raid_touch_points', 'raider_bonus',
                'raider_id', 'raiding_team_id', 'reason', 'review.outcome.id',
                'review.outcome.outcome_value', 'review.reason.id',
                'review.reason.reason_value', 'review.review_taken', 'review.tea
m_id',
                'score', 'status_id', 'substituted_by', 'super_raid', 'super_tac
kle',
                'team_id'],
                dtype='object')
```

```
In [81... events.shape
```

```
Out[81]: (66568, 37)
```

```
In [82... #Handling the missing values
```

```
In [83... events.isnull().sum()
```

```
Out[83]: Column1          0
         clock            0
         defender_id      52513
         defending_all_out_points 46769
         defending_bonus_points 46769
         defending_capture_points 46769
         defending_points    8253
         defending_team_id   8253
         defending_technical_points 46769
         do_or_die         46271
         event             0
         event_id          0
         event_no          0
         event_text        0
         match_id          0
         player_id        61127
         raid_all_out_points 46769
         raid_bonus_points 46769
         raid_points       8253
         raid_technical_points 46769
         raid_touch_points 46769
         raider_bonus      66070
         raider_id         8253
         raiding_team_id   8253
         reason            65646
         review.outcome.id 66125
         review.outcome.outcome_value 66125
         review.reason.id  66125
         review.reason.reason_value 66125
         review.review_taken 46769
         review.team_id    46769
         score             7335
         status_id         21456
         substituted_by    61835
         super_raid        46769
         super_tackle      46769
         team_id           58318
         dtype: int64
```

```
In [84... #select all the columns that ends with points
         # x="hello"
         # print(x.endswith("p"))
```

```
In [85... points_column=[col for col in events.columns if col.endswith("points")]
```

```
In [86... points_column
```

```
Out[86]: ['defending_all_out_points',  
         'defending_bonus_points',  
         'defending_capture_points',  
         'defending_points',  
         'defending_technical_points',  
         'raid_all_out_points',  
         'raid_bonus_points',  
         'raid_points',  
         'raid_technical_points',  
         'raid_touch_points']
```

```
In [87... p=[]  
for col in events.columns:  
    if col.endswith("points"):  
        p.append(col)  
print(p)
```

```
['defending_all_out_points', 'defending_bonus_points', 'defending_capture_points', 'defending_points', 'defending_technical_points', 'raid_all_out_points', 'raid_bonus_points', 'raid_points', 'raid_technical_points', 'raid_touch_points']
```

```
In [88... events[points_column]=events[points_column].fillna(0)
```

```
In [89... events.isnull().sum()
```

```

Out[89]: Column1          0
        clock            0
        defender_id      52513
        defending_all_out_points  0
        defending_bonus_points  0
        defending_capture_points  0
        defending_points    0
        defending_team_id    8253
        defending_technical_points  0
        do_or_die         46271
        event            0
        event_id         0
        event_no         0
        event_text       0
        match_id         0
        player_id        61127
        raid_all_out_points  0
        raid_bonus_points  0
        raid_points      0
        raid_technical_points  0
        raid_touch_points  0
        raider_bonus     66070
        raider_id        8253
        raiding_team_id   8253
        reason           65646
        review.outcome.id 66125
        review.outcome.outcome_value 66125
        review.reason.id  66125
        review.reason.reason_value 66125
        review.review_taken 46769
        review.team_id    46769
        score            7335
        status_id        21456
        substituted_by    61835
        super_raid        46769
        super_tackle      46769
        team_id          58318
        dtype: int64

```

```

In [90... #ends with id
id_column=[col for col in events.columns if col.endswith("_id") or col.en

```

```

In [91... id_column

```

```
Out[91]: ['defender_id',
          'defending_team_id',
          'event_id',
          'match_id',
          'player_id',
          'raider_id',
          'raiding_team_id',
          'review.outcome.id',
          'review.reason.id',
          'review.team_id',
          'status_id',
          'team_id']
```

```
In [92... #replace -
events[id_column]=events[id_column].fillna("-")
```

```
In [93... events[id_column].isnull().sum()
```

```
Out[93]: defender_id      0
          defending_team_id  0
          event_id         0
          match_id         0
          player_id        0
          raider_id        0
          raiding_team_id   0
          review.outcome.id  0
          review.reason.id   0
          review.team_id    0
          status_id         0
          team_id           0
          dtype: int64
```

```
In [94... events.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 66568 entries, 0 to 66567
```

```
Data columns (total 37 columns):
```

#	Column	Non-Null Count	Dtype
0	Column1	66568 non-null	int64
1	clock	66568 non-null	category
2	defender_id	66568 non-null	object
3	defending_all_out_points	66568 non-null	float64
4	defending_bonus_points	66568 non-null	float64
5	defending_capture_points	66568 non-null	float64
6	defending_points	66568 non-null	float64
7	defending_team_id	66568 non-null	object
8	defending_technical_points	66568 non-null	float64
9	do_or_die	20297 non-null	category
10	event	66568 non-null	category
11	event_id	66568 non-null	int64
12	event_no	66568 non-null	int64
13	event_text	66568 non-null	category
14	match_id	66568 non-null	int64
15	player_id	66568 non-null	object
16	raid_all_out_points	66568 non-null	float64
17	raid_bonus_points	66568 non-null	float64
18	raid_points	66568 non-null	float64
19	raid_technical_points	66568 non-null	float64
20	raid_touch_points	66568 non-null	float64
21	raider_bonus	498 non-null	float64
22	raider_id	66568 non-null	object
23	raiding_team_id	66568 non-null	object
24	reason	922 non-null	category
25	review.outcome.id	66568 non-null	object
26	review.outcome.outcome_value	443 non-null	category
27	review.reason.id	66568 non-null	object
28	review.reason.reason_value	443 non-null	category
29	review.review_taken	19799 non-null	category
30	review.team_id	66568 non-null	object
31	score	59233 non-null	category
32	status_id	66568 non-null	object
33	substituted_by	4733 non-null	float64
34	super_raid	19799 non-null	float64
35	super_tackle	19799 non-null	float64
36	team_id	66568 non-null	object

```
dtypes: category(9), float64(14), int64(4), object(10)
```

```
memory usage: 15.4+ MB
```

```
In [95... #numerical data
#null--->median
numerical_columns=events.select_dtypes(include=[np.number]).columns
```

```
In [96... numerical_columns
```

```
Out[96]: Index(['Column1', 'defending_all_out_points', 'defending_bonus_points',
               'defending_capture_points', 'defending_points',
               'defending_technical_points', 'event_id', 'event_no', 'match_i
               d',
               'raid_all_out_points', 'raid_bonus_points', 'raid_points',
               'raid_technical_points', 'raid_touch_points', 'raider_bonus',
               'substituted_by', 'super_raid', 'super_tackle'],
              dtype='object')
```

```
In [97... for column in numerical_columns:
          events[column].fillna(events[column].median(), inplace=True)
```

C:\Users\Deeksha\AppData\Local\Temp\ipykernel_26676\2539149993.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
events[column].fillna(events[column].median(), inplace=True)
```

```
In [98... events[numerical_columns].isnull().sum()
```

```
Out[98]: Column1                                0
defending_all_out_points                      0
defending_bonus_points                       0
defending_capture_points                     0
defending_points                             0
defending_technical_points                   0
event_id                                     0
event_no                                     0
match_id                                    0
raid_all_out_points                          0
raid_bonus_points                           0
raid_points                                 0
raid_technical_points                       0
raid_touch_points                           0
raider_bonus                                0
substituted_by                              0
super_raid                                  0
super_tackle                                0
dtype: int64
```

```
In [99... events.info()
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 66568 entries, 0 to 66567

Data columns (total 37 columns):

#	Column	Non-Null Count	Dtype
0	Column1	66568 non-null	int64
1	clock	66568 non-null	category
2	defender_id	66568 non-null	object
3	defending_all_out_points	66568 non-null	float64
4	defending_bonus_points	66568 non-null	float64
5	defending_capture_points	66568 non-null	float64
6	defending_points	66568 non-null	float64
7	defending_team_id	66568 non-null	object
8	defending_technical_points	66568 non-null	float64
9	do_or_die	20297 non-null	category
10	event	66568 non-null	category
11	event_id	66568 non-null	int64
12	event_no	66568 non-null	int64
13	event_text	66568 non-null	category
14	match_id	66568 non-null	int64
15	player_id	66568 non-null	object
16	raid_all_out_points	66568 non-null	float64
17	raid_bonus_points	66568 non-null	float64
18	raid_points	66568 non-null	float64
19	raid_technical_points	66568 non-null	float64
20	raid_touch_points	66568 non-null	float64
21	raider_bonus	66568 non-null	float64
22	raider_id	66568 non-null	object
23	raiding_team_id	66568 non-null	object
24	reason	922 non-null	category
25	review.outcome.id	66568 non-null	object
26	review.outcome.outcome_value	443 non-null	category
27	review.reason.id	66568 non-null	object
28	review.reason.reason_value	443 non-null	category
29	review.review_taken	19799 non-null	category
30	review.team_id	66568 non-null	object
31	score	59233 non-null	category
32	status_id	66568 non-null	object
33	substituted_by	66568 non-null	float64
34	super_raid	66568 non-null	float64
35	super_tackle	66568 non-null	float64
36	team_id	66568 non-null	object

dtypes: category(9), float64(14), int64(4), object(10)

memory usage: 15.4+ MB

```
In [10...] categorical_columns=events.select_dtypes(include=["category"]).columns
```

```
In [10...] categorical_columns
```

```
Out[101]: Index(['clock', 'do_or_die', 'event', 'event_text', 'reason',  
              'review.outcome.outcome_value', 'review.reason.reason_value',  
              'review.review_taken', 'score'],  
              dtype='object')
```

```
In [10... for column in categorical_columns:
          events[column].fillna(events[column].mode()[0], inplace=True)
```

C:\Users\Deeksha\AppData\Local\Temp\ipykernel_26676\744490760.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
events[column].fillna(events[column].mode()[0], inplace=True)
```

```
In [10... events.isnull().sum()
```

```

Out[103]: Column1      0
          clock        0
          defender_id  0
          defending_all_out_points  0
          defending_bonus_points  0
          defending_capture_points  0
          defending_points  0
          defending_team_id  0
          defending_technical_points  0
          do_or_die      0
          event          0
          event_id      0
          event_no      0
          event_text     0
          match_id      0
          player_id     0
          raid_all_out_points  0
          raid_bonus_points  0
          raid_points    0
          raid_technical_points  0
          raid_touch_points  0
          raider_bonus   0
          raider_id     0
          raiding_team_id  0
          reason        0
          review.outcome.id  0
          review.outcome.outcome_value  0
          review.reason.id  0
          review.reason.reason_value  0
          review.review_taken  0
          review.team_id  0
          score         0
          status_id     0
          substituted_by  0
          super_raid     0
          super_tackle   0
          team_id       0
          dtype: int64

```

```

In [10... #ensure the data consistency in numerical columns
          #points--->-ve
          p=["defending_points","raid_points"]

```

```

In [10... for c in p:
          events[c]=events[c].apply(lambda x:max(x,0))

```

```

In [ ]: # max(-1,0)

```

```

In [ ]: # events["defending_points"]=events["defending_points"].apply(lambda x:ma

```

```

In [10... events.info()

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 66568 entries, 0 to 66567
```

```
Data columns (total 37 columns):
```

#	Column	Non-Null Count	Dtype
0	Column1	66568 non-null	int64
1	clock	66568 non-null	category
2	defender_id	66568 non-null	object
3	defending_all_out_points	66568 non-null	float64
4	defending_bonus_points	66568 non-null	float64
5	defending_capture_points	66568 non-null	float64
6	defending_points	66568 non-null	float64
7	defending_team_id	66568 non-null	object
8	defending_technical_points	66568 non-null	float64
9	do_or_die	66568 non-null	category
10	event	66568 non-null	category
11	event_id	66568 non-null	int64
12	event_no	66568 non-null	int64
13	event_text	66568 non-null	category
14	match_id	66568 non-null	int64
15	player_id	66568 non-null	object
16	raid_all_out_points	66568 non-null	float64
17	raid_bonus_points	66568 non-null	float64
18	raid_points	66568 non-null	float64
19	raid_technical_points	66568 non-null	float64
20	raid_touch_points	66568 non-null	float64
21	raider_bonus	66568 non-null	float64
22	raider_id	66568 non-null	object
23	raiding_team_id	66568 non-null	object
24	reason	66568 non-null	category
25	review.outcome.id	66568 non-null	object
26	review.outcome.outcome_value	66568 non-null	category
27	review.reason.id	66568 non-null	object
28	review.reason.reason_value	66568 non-null	category
29	review.review_taken	66568 non-null	category
30	review.team_id	66568 non-null	object
31	score	66568 non-null	category
32	status_id	66568 non-null	object
33	substituted_by	66568 non-null	float64
34	super_raid	66568 non-null	float64
35	super_tackle	66568 non-null	float64
36	team_id	66568 non-null	object

```
dtypes: category(9), float64(14), int64(4), object(10)
```

```
memory usage: 15.4+ MB
```

```
In [10... events.head()
```

```
Out[108]:
```

	Column1	clock	defender_id	defending_all_out_points	defending_bonus_points	defending
0	0	0:33	-	0.0	0.0	
1	1	0:43	-	0.0	0.0	
2	2	0:48	54.0	0.0	0.0	
3	3	1:42	-	0.0	0.0	
4	4	2:19	-	0.0	0.0	

5 rows × 37 columns

```
In [10... #ensure the data consistency in categorical columns
events["event"].value_counts()
```

```
Out[109]: event
Empty Raid                24951
Successful Raid            20218
Unsuccessful Raid         13119
Substitution               4733
TimeOut                   2588
Green Card                 657
Non Raid Technical Points  224
Yellow Card                48
Raider and Anti Out       27
Red Card                   3
Name: count, dtype: int64
```

```
In [11... events["event_text"].value_counts()
```

```
Out[110]: event_text
Timeout taken by Umpire                1616
Pardeep Narwal raids successfully      877
Rahul Chaudhari Empty Raid             811
Anup Kumar Empty Raid                  764
Rahul Chaudhari raids successfully     706
...
Gurvinder Singh comes in for Rakesh Kumar    1
Rajesh Dhiman comes in for Mofimondal        1
Rajesh Manokaran comes in for Ajay Thakur    1
Gurvinder Singh comes in for Mahesh Goud    1
S. M. Al Mamun comes in for Jahangir Alam    1
Name: count, Length: 5413, dtype: int64
```

```
In [11... events["event"]=events["event"].str.lower().str.strip()
```

```
In [11... events["event"].value_counts()
```

```
Out[113]: event
empty raid                24951
successful raid           20218
unsuccessful raid        13119
substitution              4733
timeout                  2588
green card                657
non raid technical points  224
yellow card               48
raider and anti out       27
red card                  3
Name: count, dtype: int64
```

```
In [11... events["event_text"]=events["event_text"].str.lower().str.strip()
```

```
In [11... events["event_text"].value_counts()
```

```
Out[115]: event_text
timeout taken by umpire                1616
pardeep narwal raids successfully      877
rahul chaudhari empty raid             811
anup kumar empty raid                  764
rahul chaudhari raids successfully     706
...
surender singh comes in for pankaj      1
abhishek singh comes in for rajaguru subramanian 1
rakesh kumar comes in for munish        1
munish comes in for rakesh kumar        1
deepak kumar comes in for mahendra ganesh rajput 1
Name: count, Length: 5412, dtype: int64
```

```
In [11... #data transformations
events.head()
```

```
Out[116]:
```

	Column1	clock	defender_id	defending_all_out_points	defending_bonus_points	defending
0	0	0:33	-	0.0	0.0	
1	1	0:43	-	0.0	0.0	
2	2	0:48	54.0	0.0	0.0	
3	3	1:42	-	0.0	0.0	
4	4	2:19	-	0.0	0.0	

5 rows x 37 columns

```
In [11... events["correct_defending_points"]=events["defending_all_out_points"]+eve
```

```
In [11... events["correct_defending_points"]
```

```
Out[118]: 0      0.0
          1      0.0
          2      0.0
          3      0.0
          4      0.0
          ...
          66563  0.0
          66564  0.0
          66565  0.0
          66566  0.0
          66567  0.0
          Name: correct_defending_points, Length: 66568, dtype: float64
```

```
In [12... events.head()
```

```
Out[120]:
```

	Column1	clock	defender_id	defending_all_out_points	defending_bonus_points	defending
0	0	0:33	-	0.0	0.0	
1	1	0:43	-	0.0	0.0	
2	2	0:48	54.0	0.0	0.0	
3	3	1:42	-	0.0	0.0	
4	4	2:19	-	0.0	0.0	

5 rows × 38 columns

```
In [12... events[events["correct_defending_points"]!=events["defending_points"]].he
```

```
Out[121]:
```

	Column1	clock	defender_id	defending_all_out_points	defending_bonus_points	defend
2	2	0:48	54.0	0.0	0.0	
3	3	1:42	-	0.0	0.0	
9	9	4:8	123.0	0.0	0.0	
15	15	7:5	-	0.0	0.0	
18	18	8:44	146.0	0.0	0.0	

5 rows × 38 columns

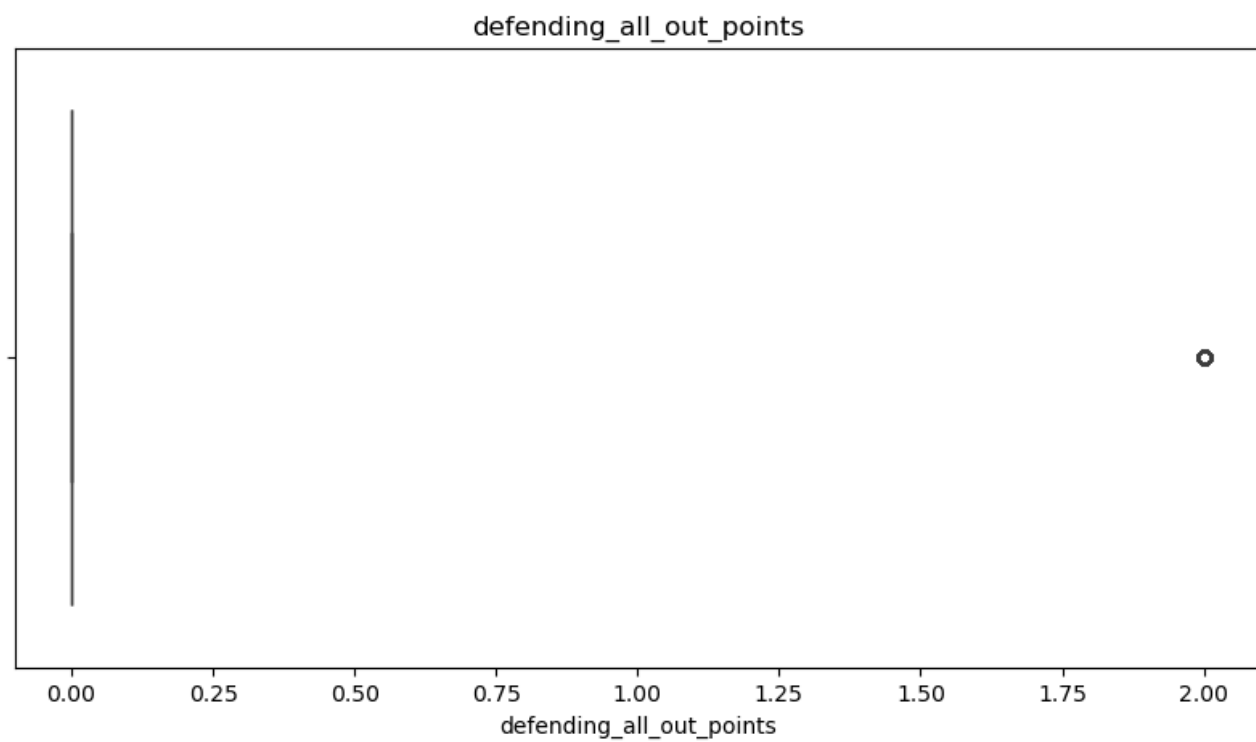
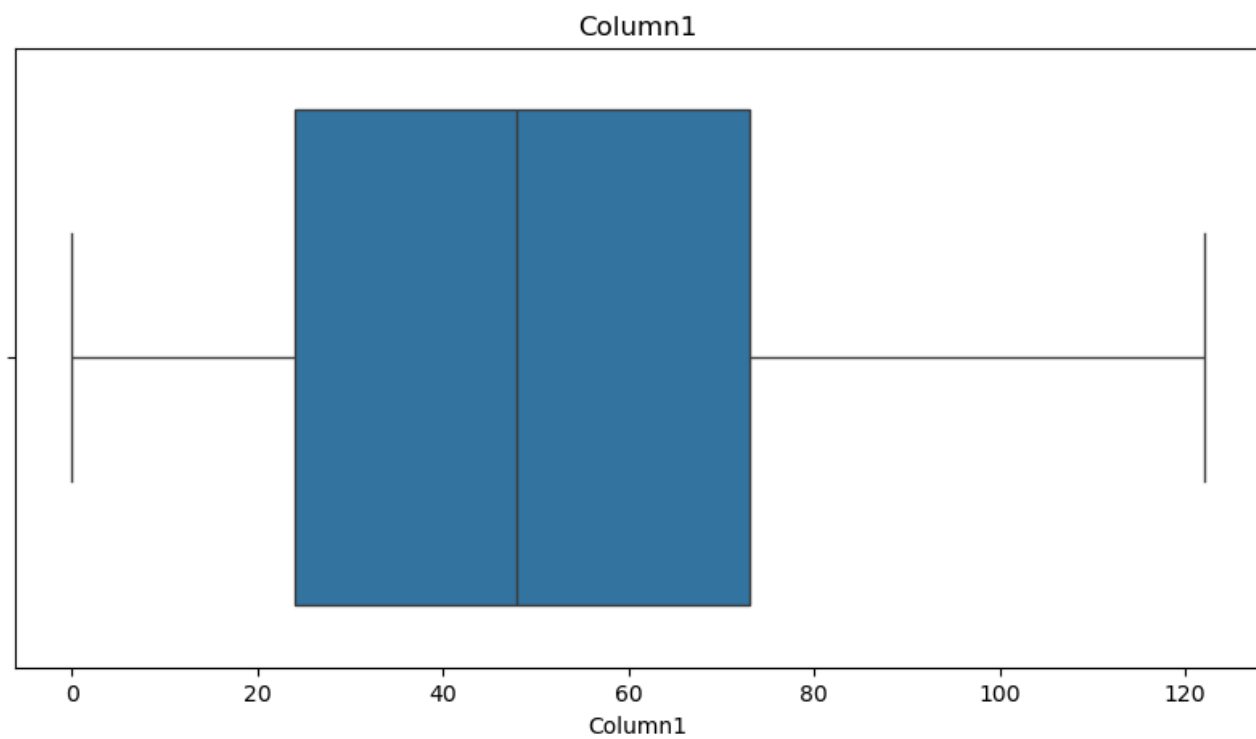
```
In [12... events[events["correct_defending_points"]!=events["defending_points"]].sh
```

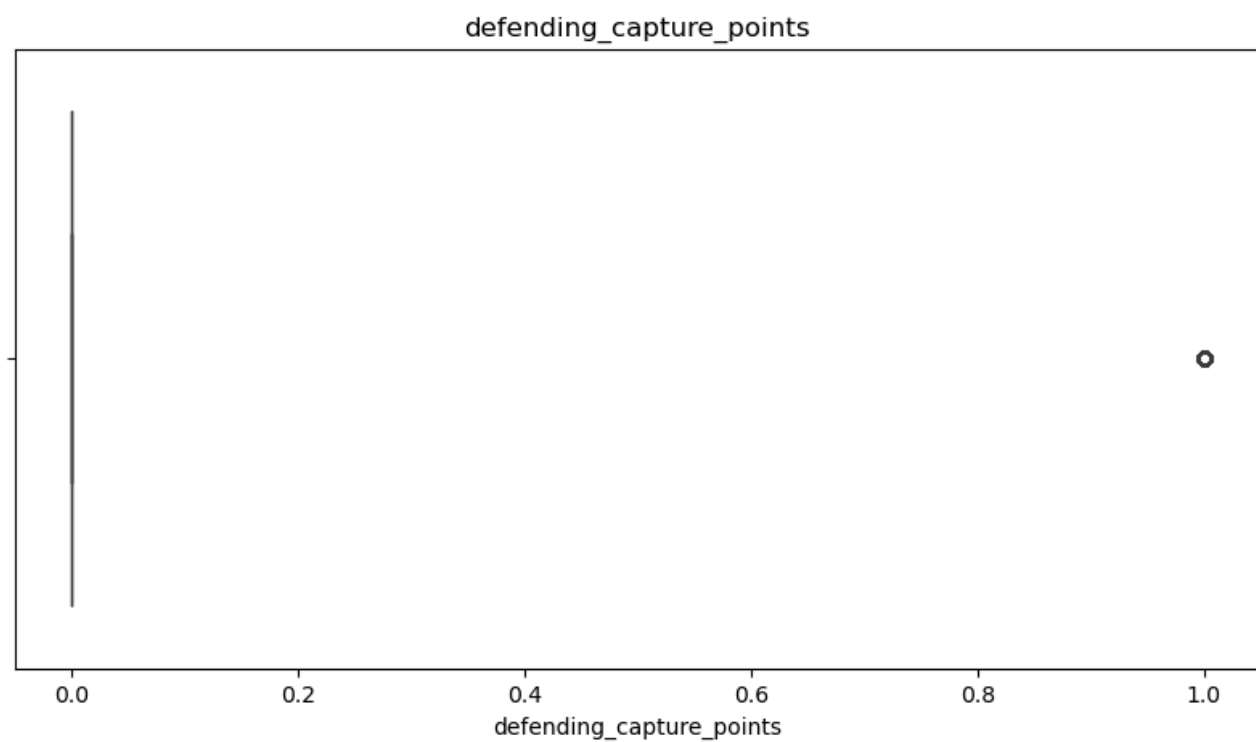
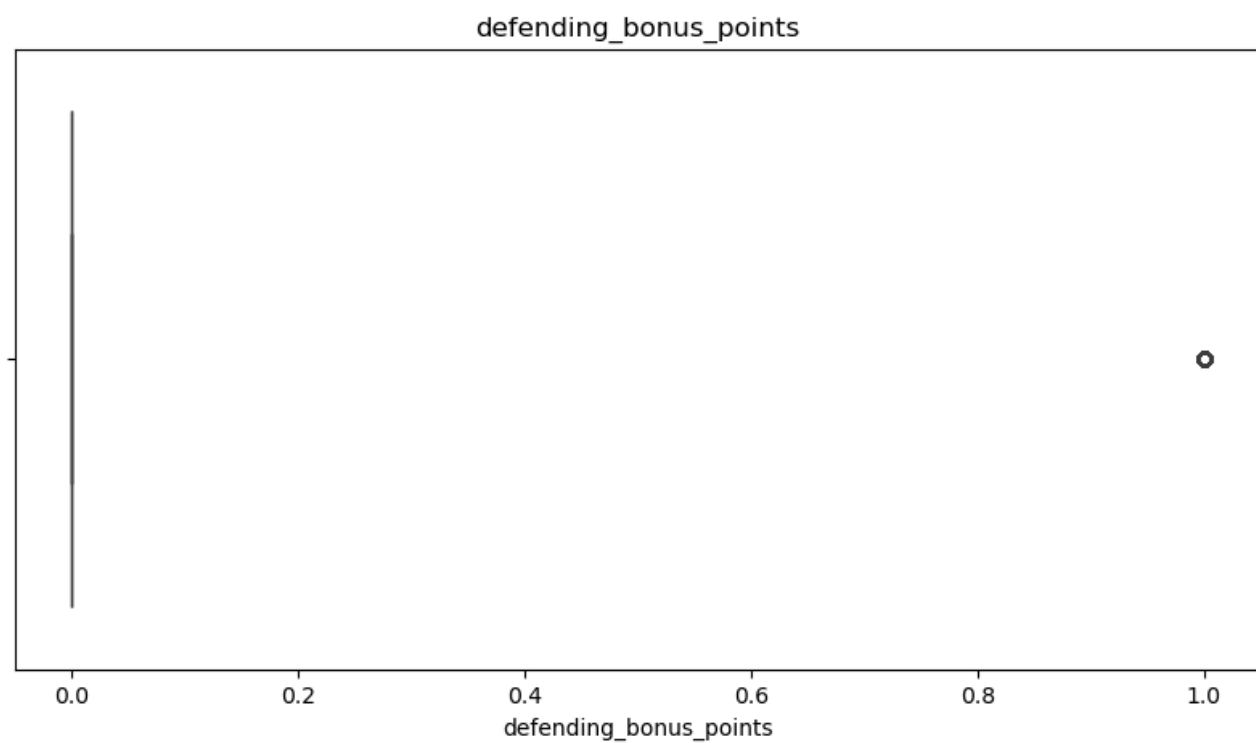
```
Out[122]: (8812, 38)
```

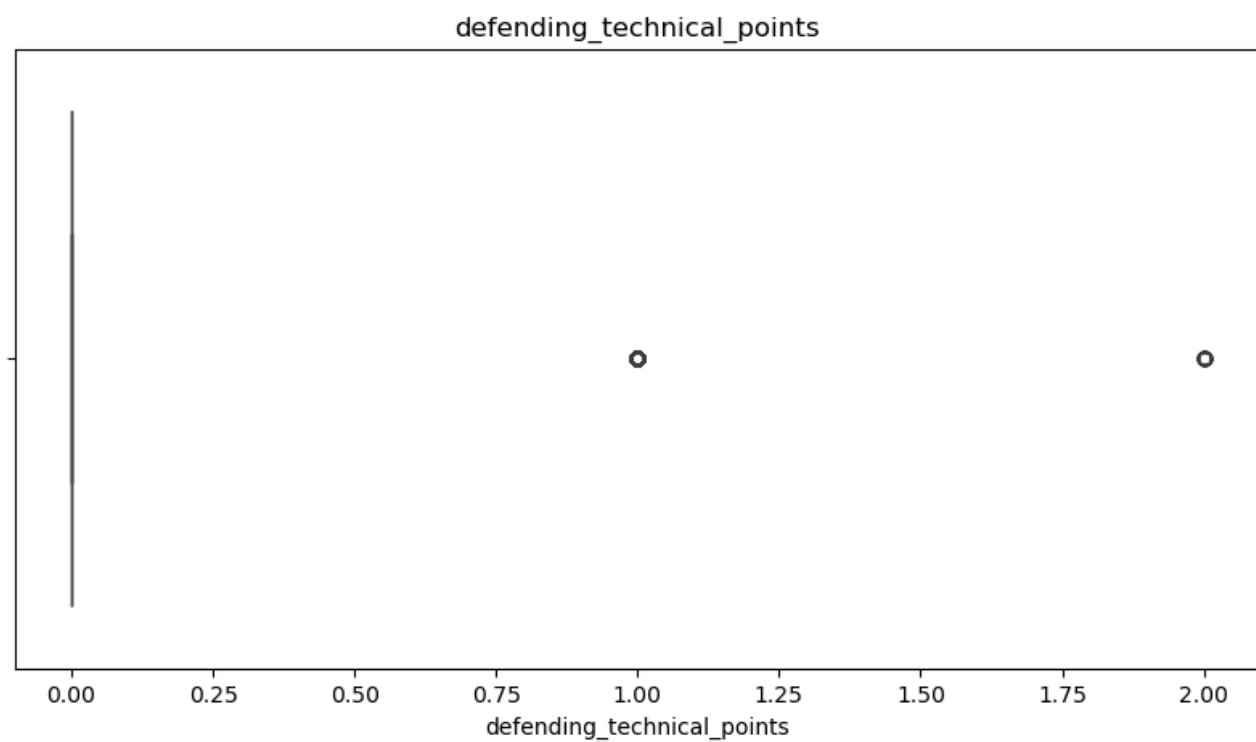
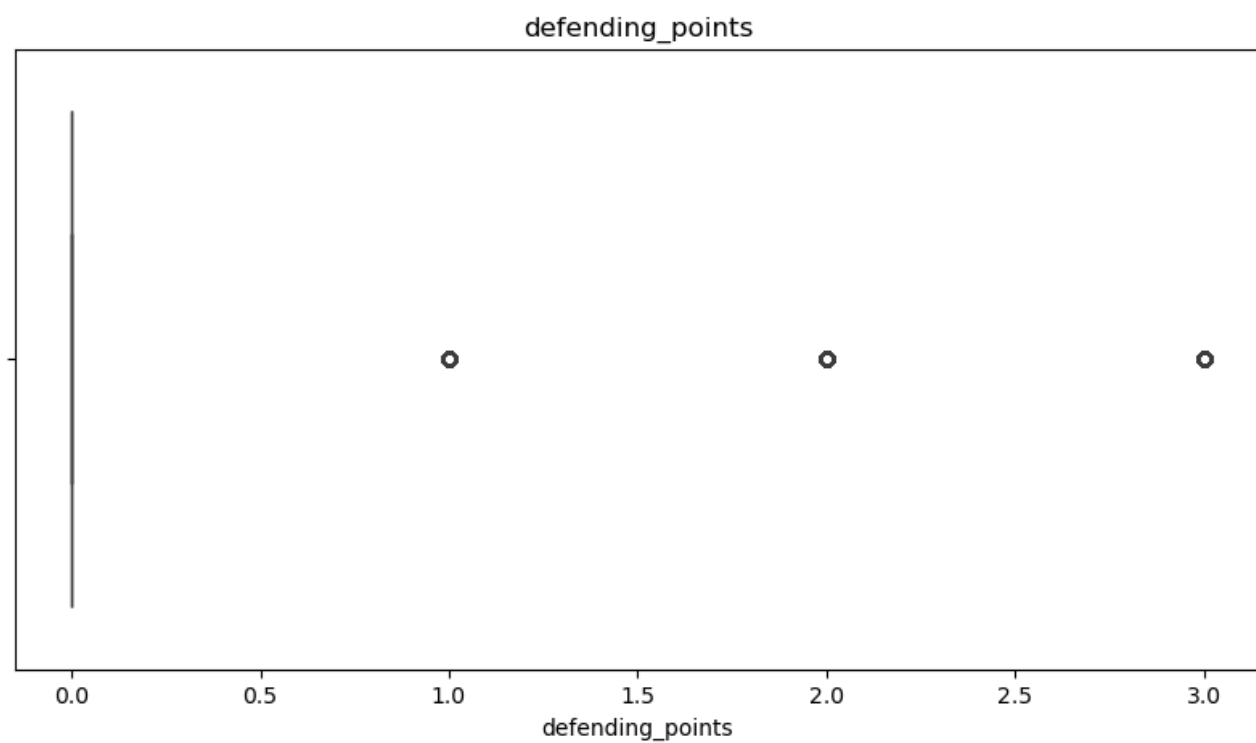
```
In [ ]: #drop defending column-->task
```

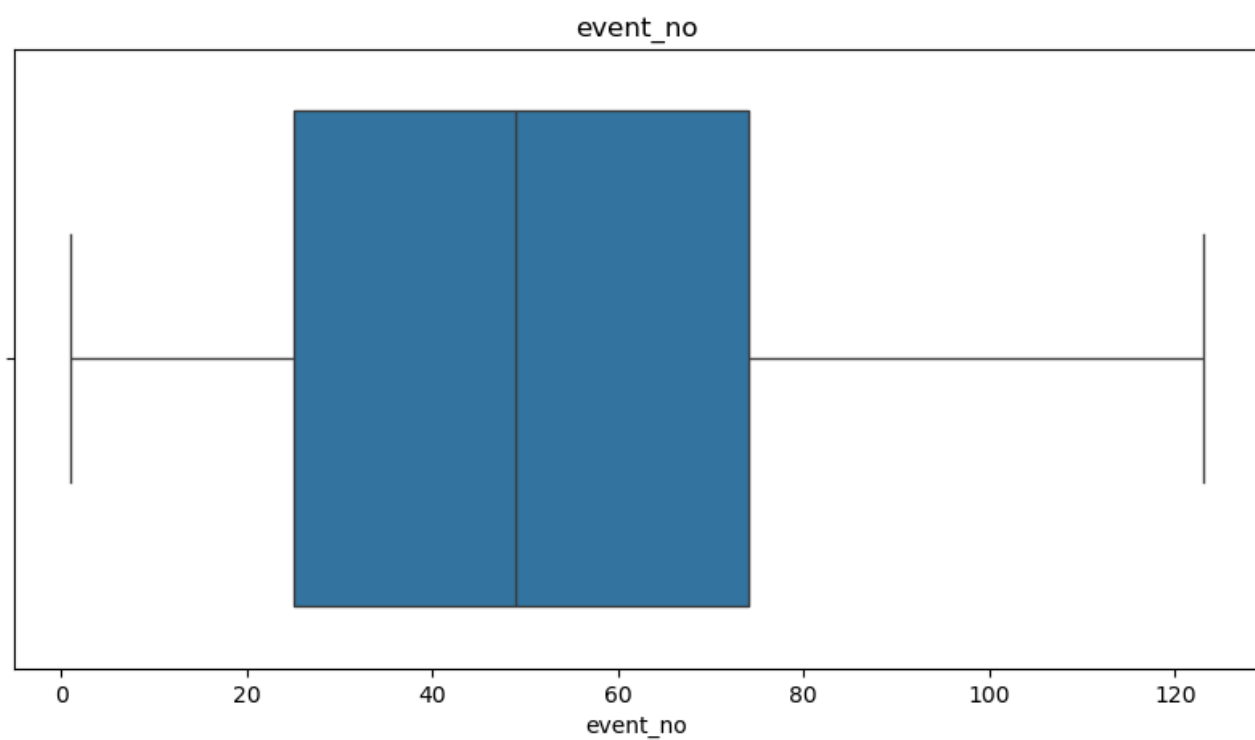
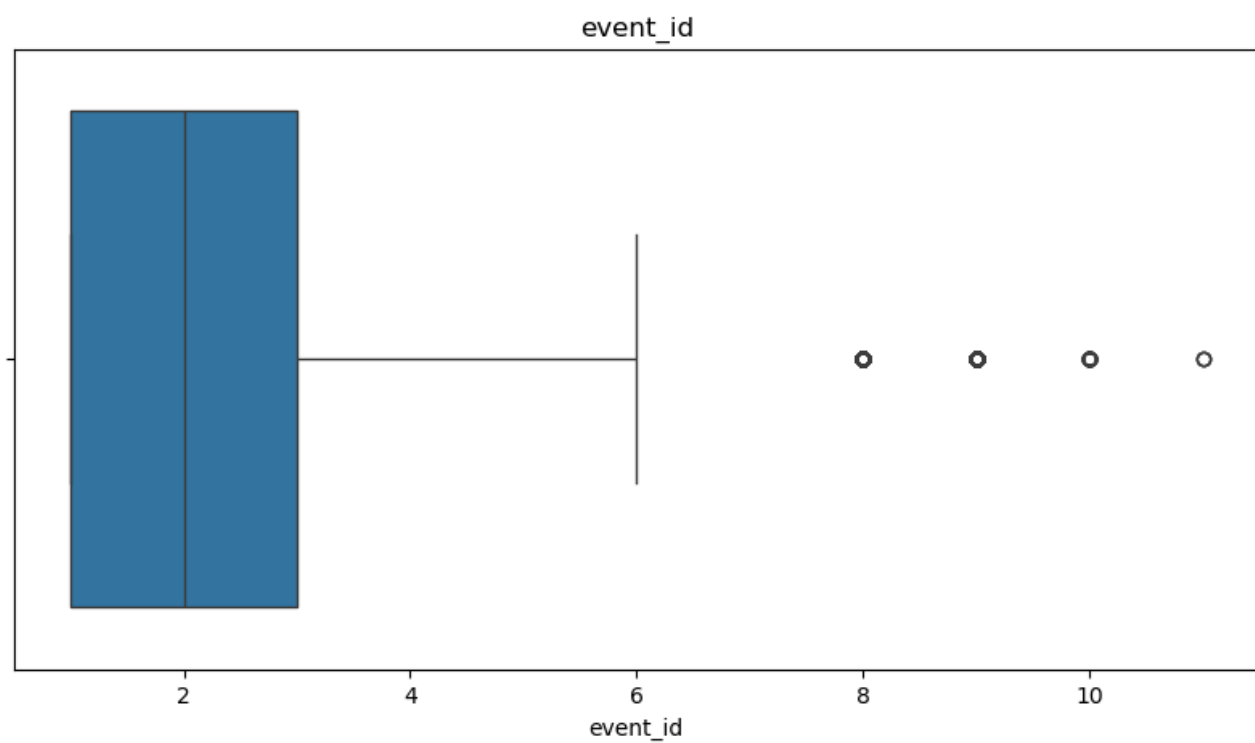
```
In [12... #identify the outliers
#all the numerical columns
#box plot
numerical_columns=events.select_dtypes(include=[np.number]).columns
```

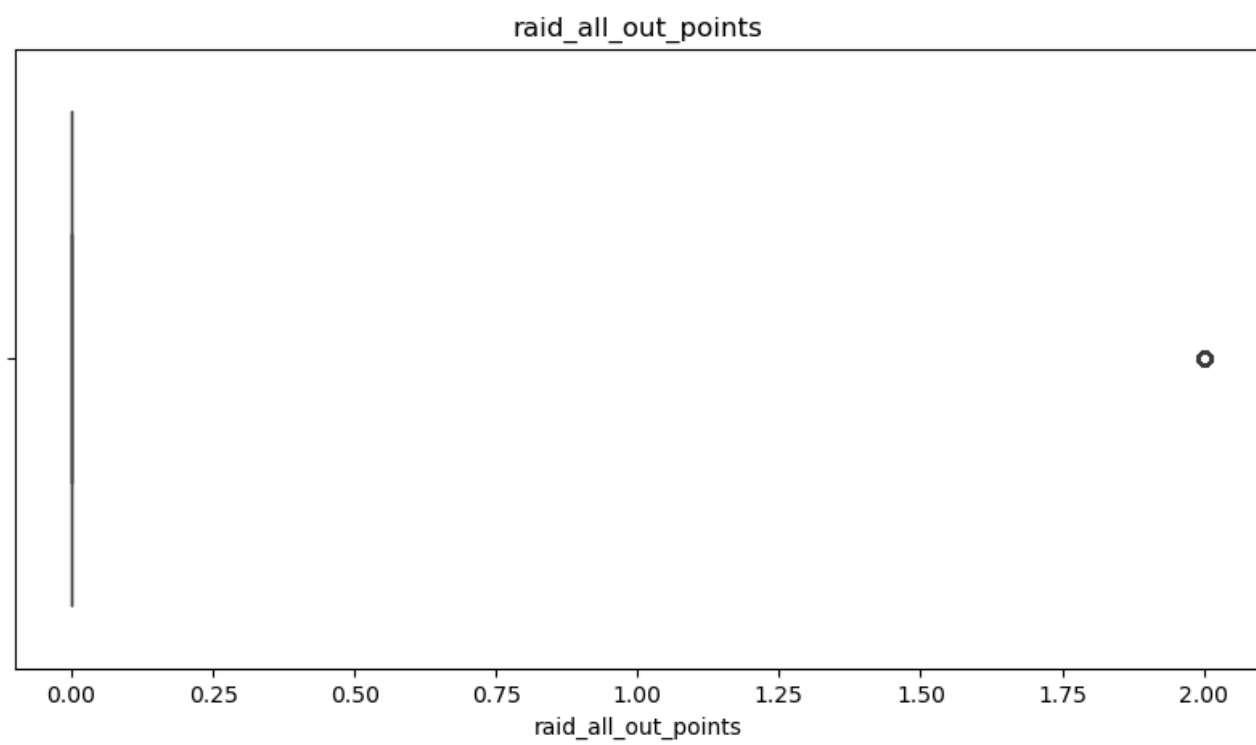
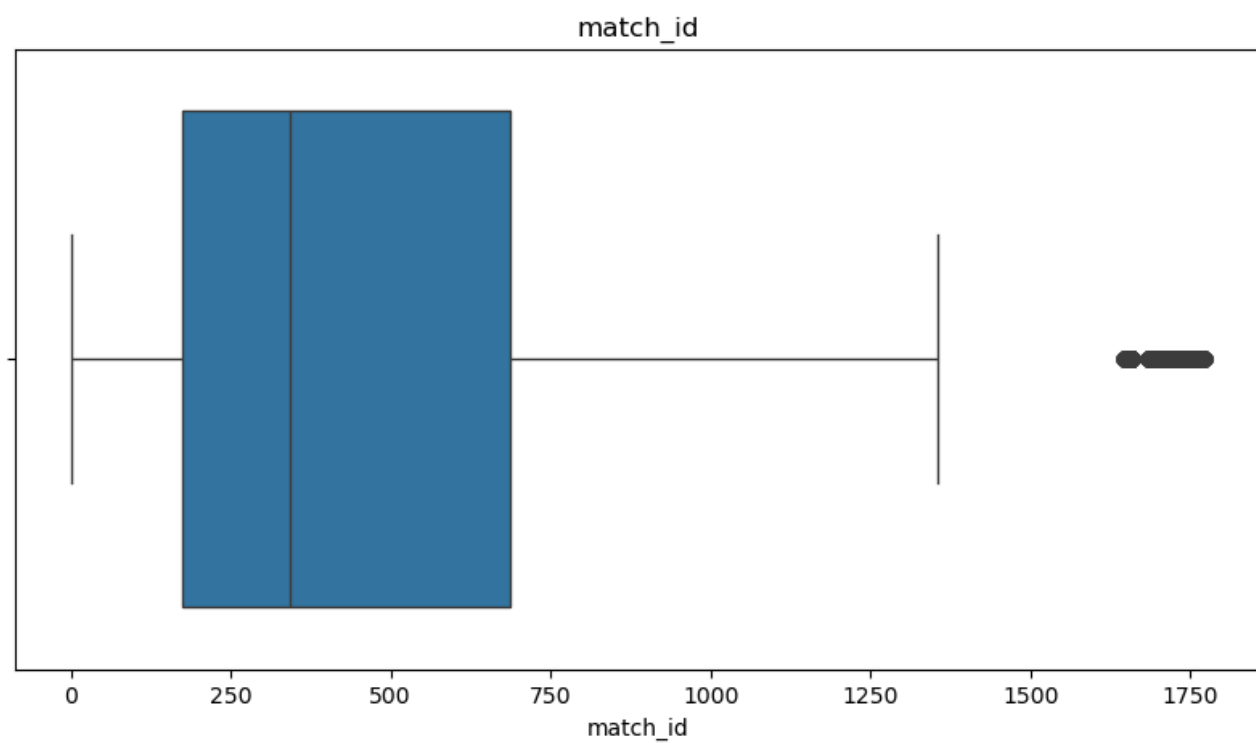
```
for column in numerical_columns:  
    plt.figure(figsize=(10,5))  
    sns.boxplot(x=events[column])  
    plt.title(column)  
    plt.show()
```

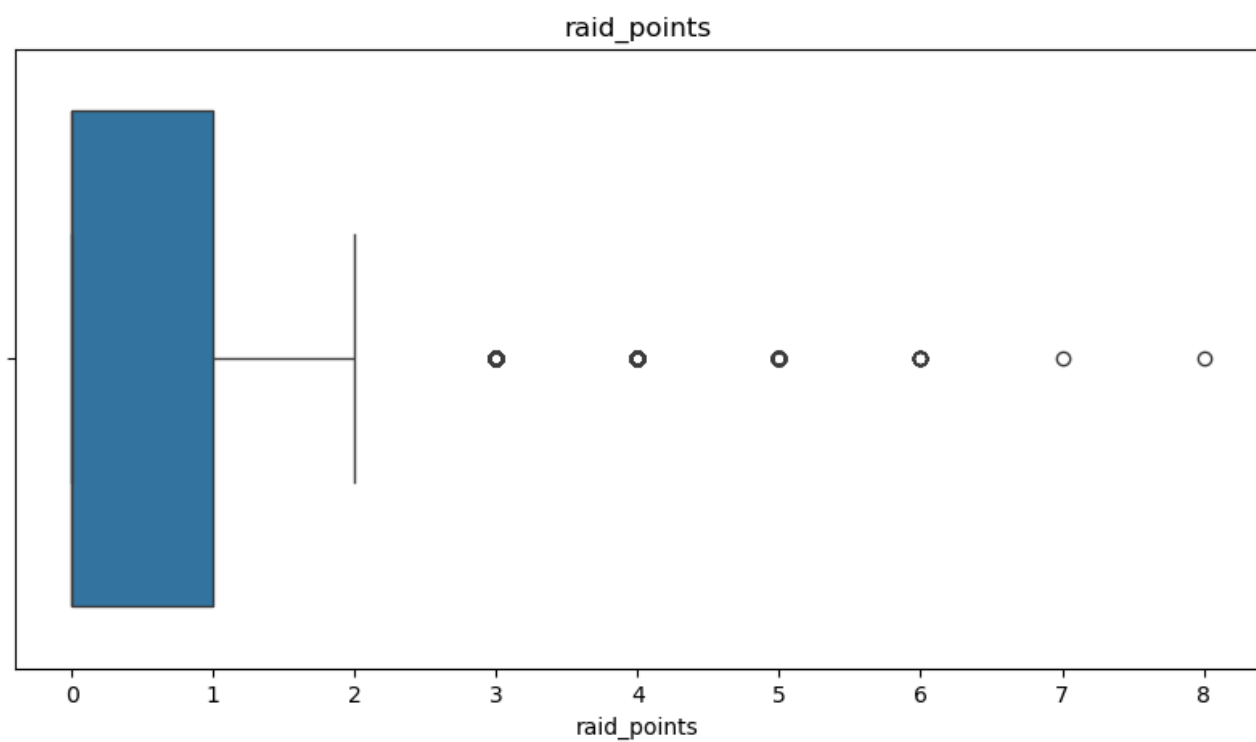
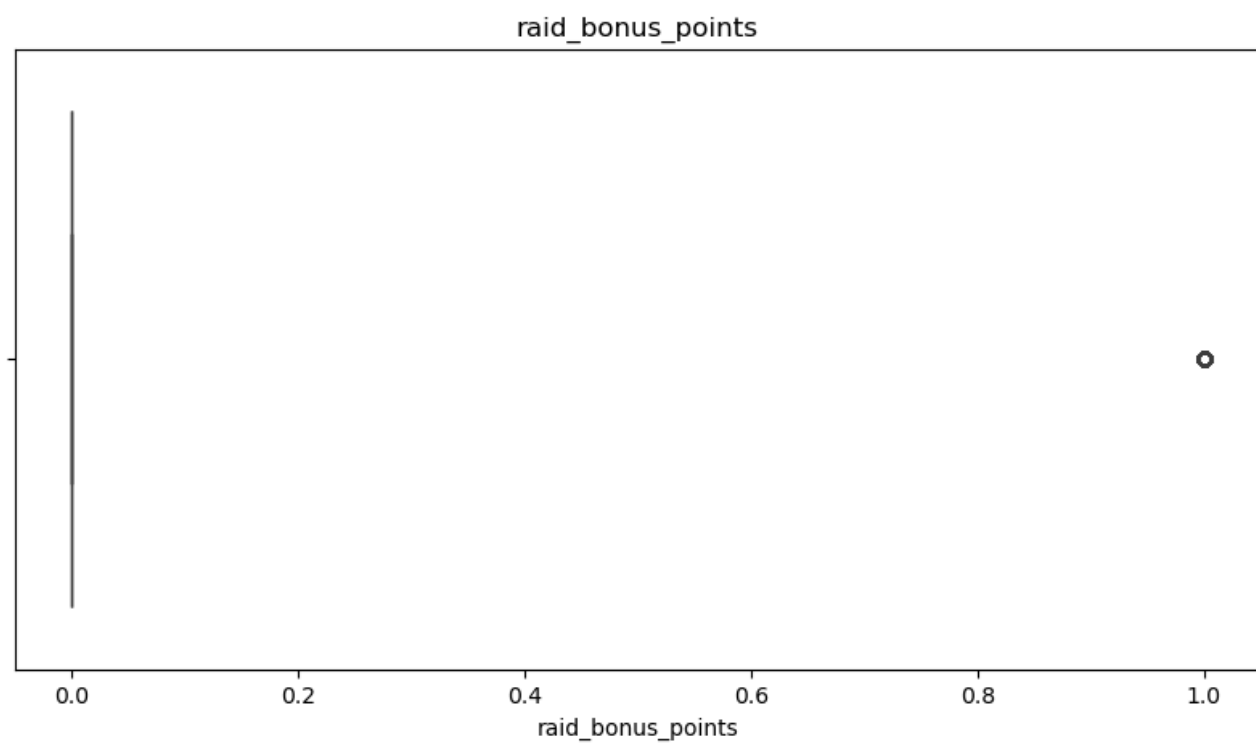


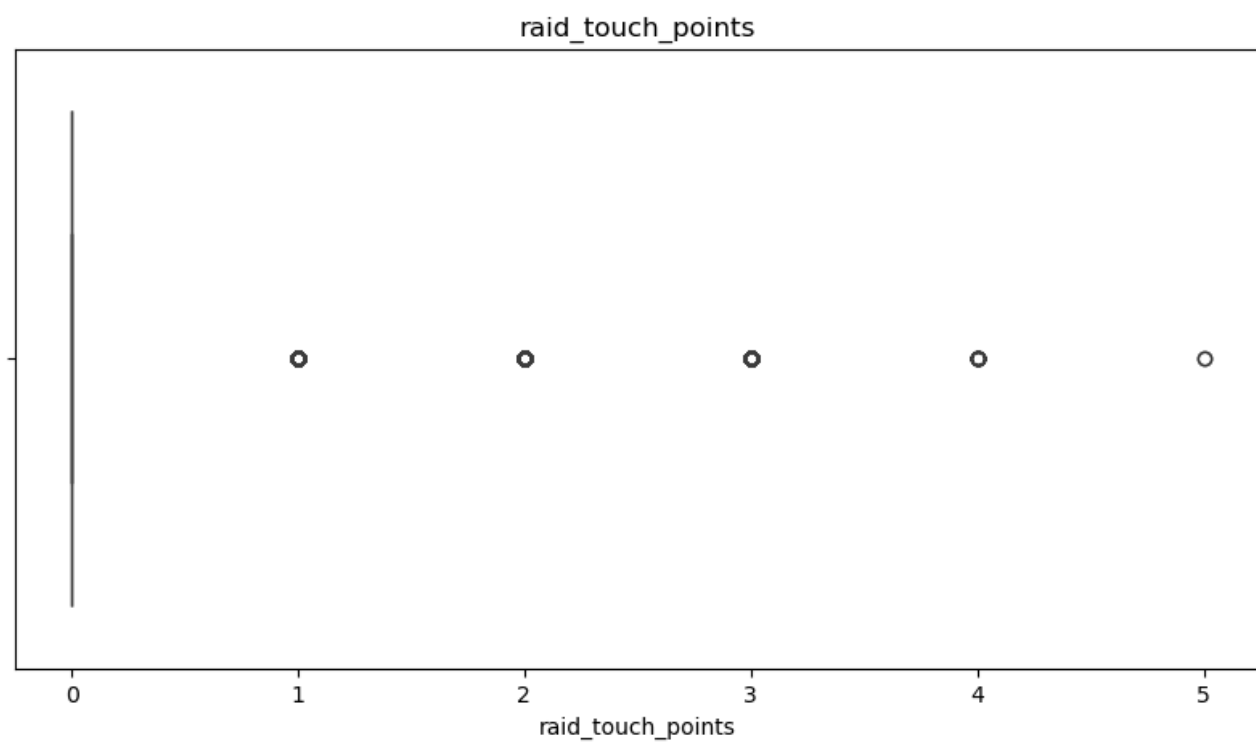
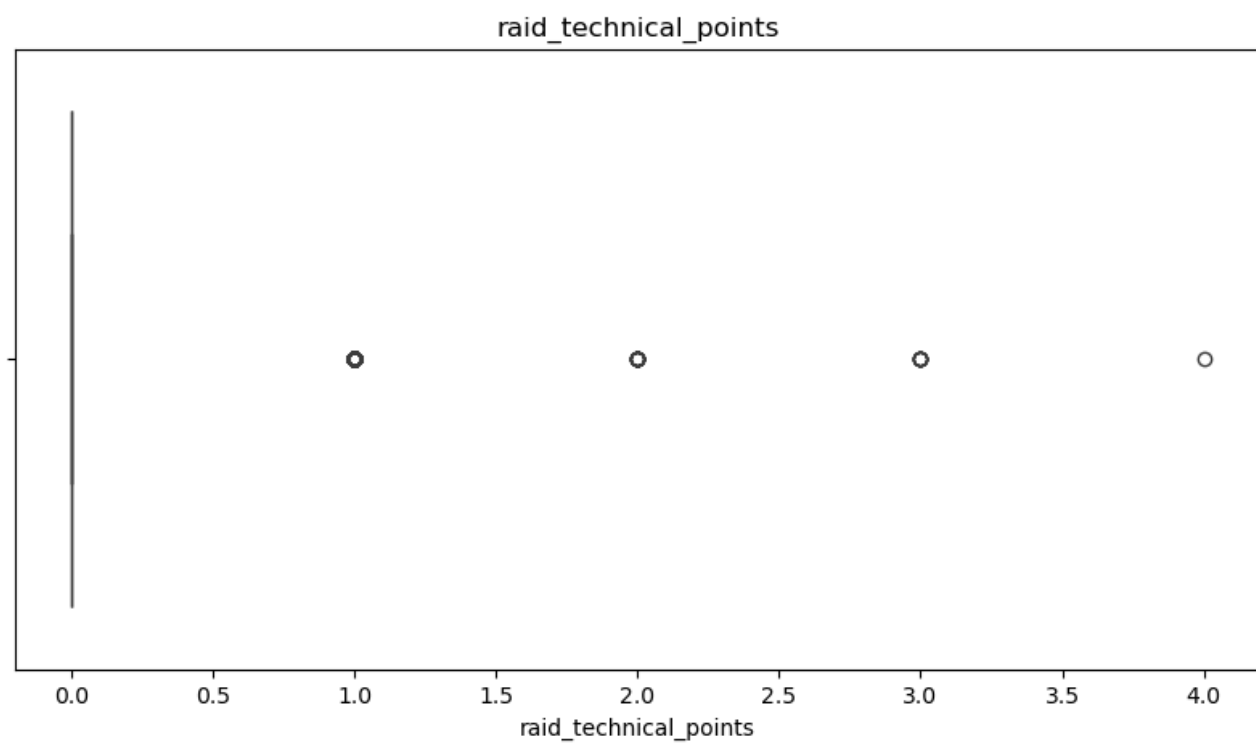


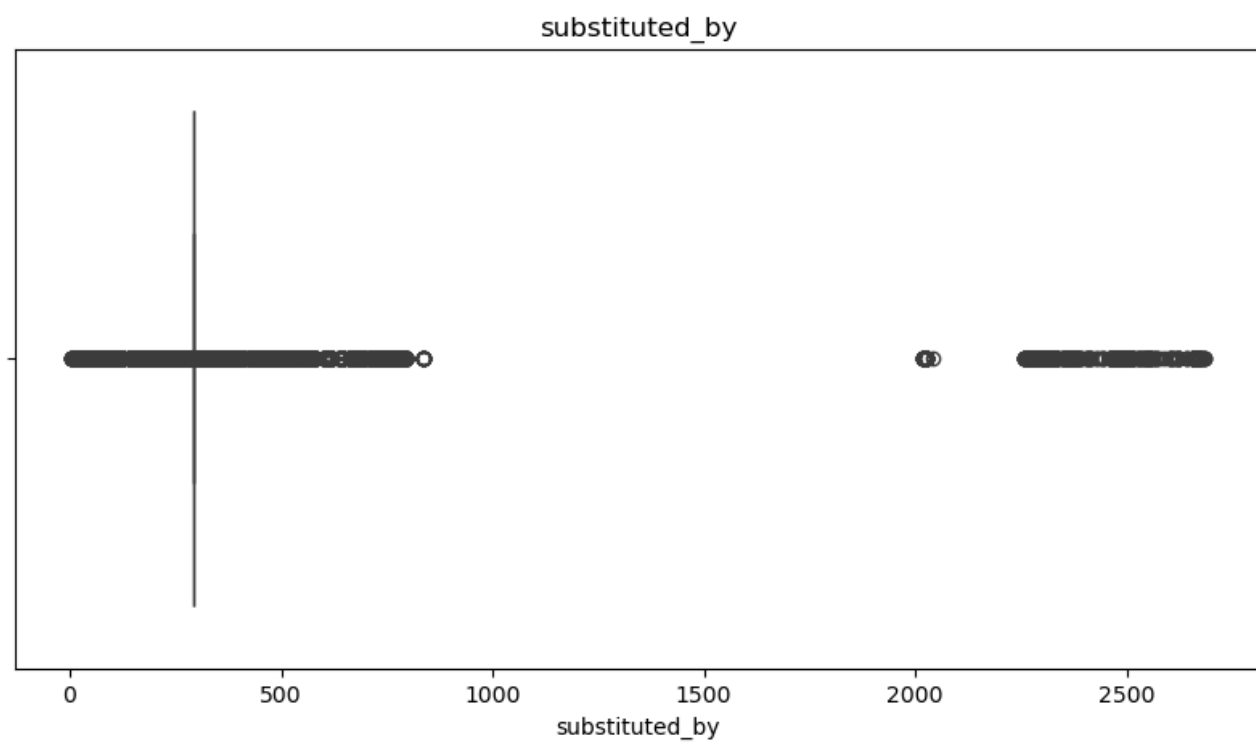
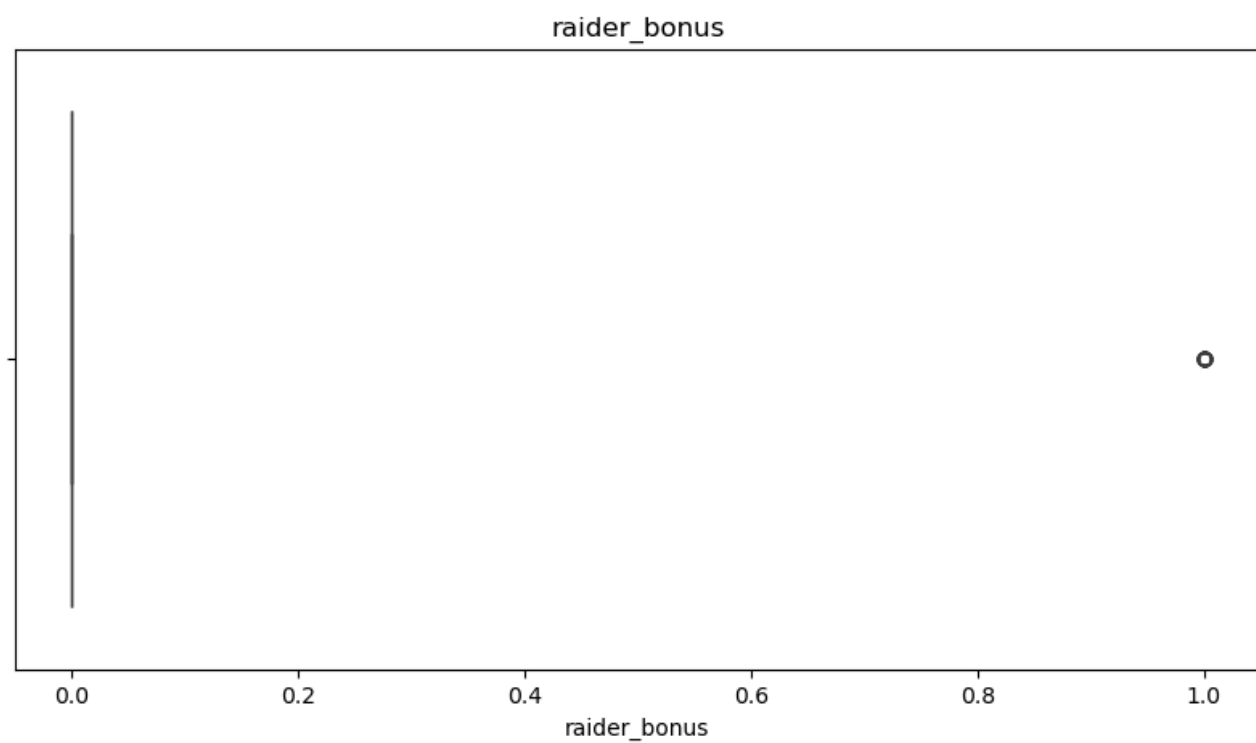


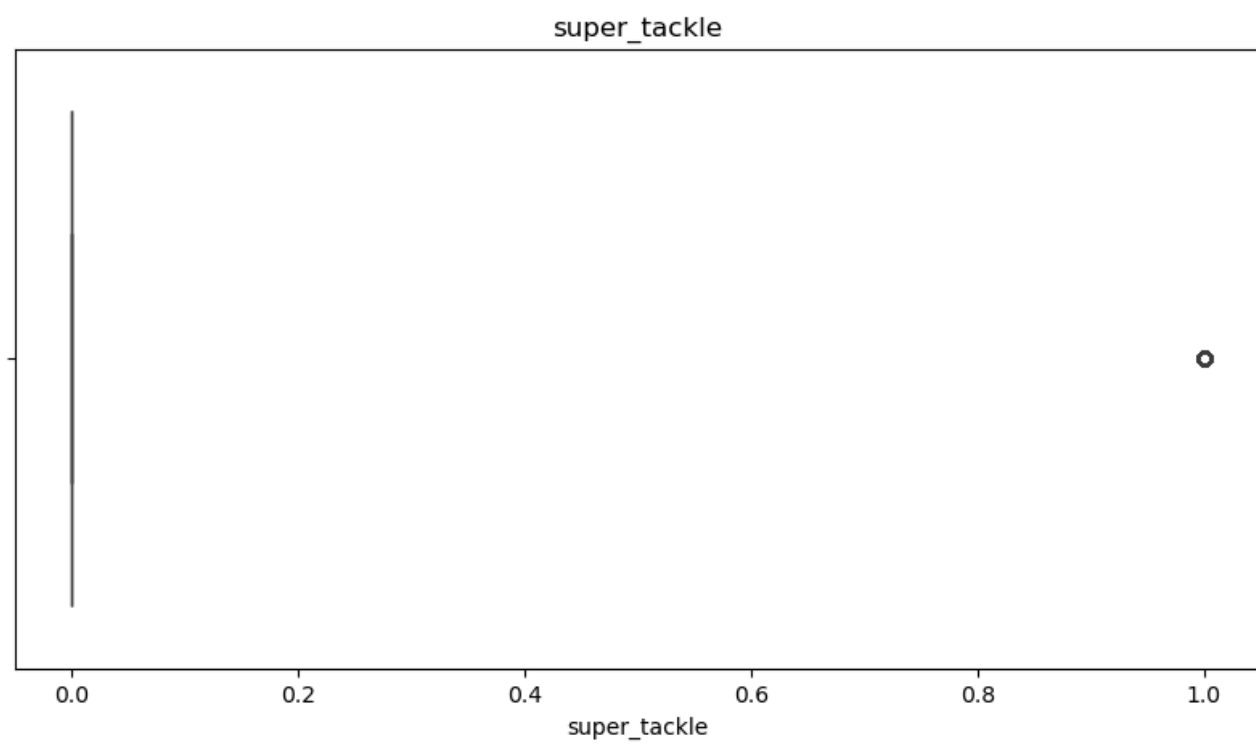
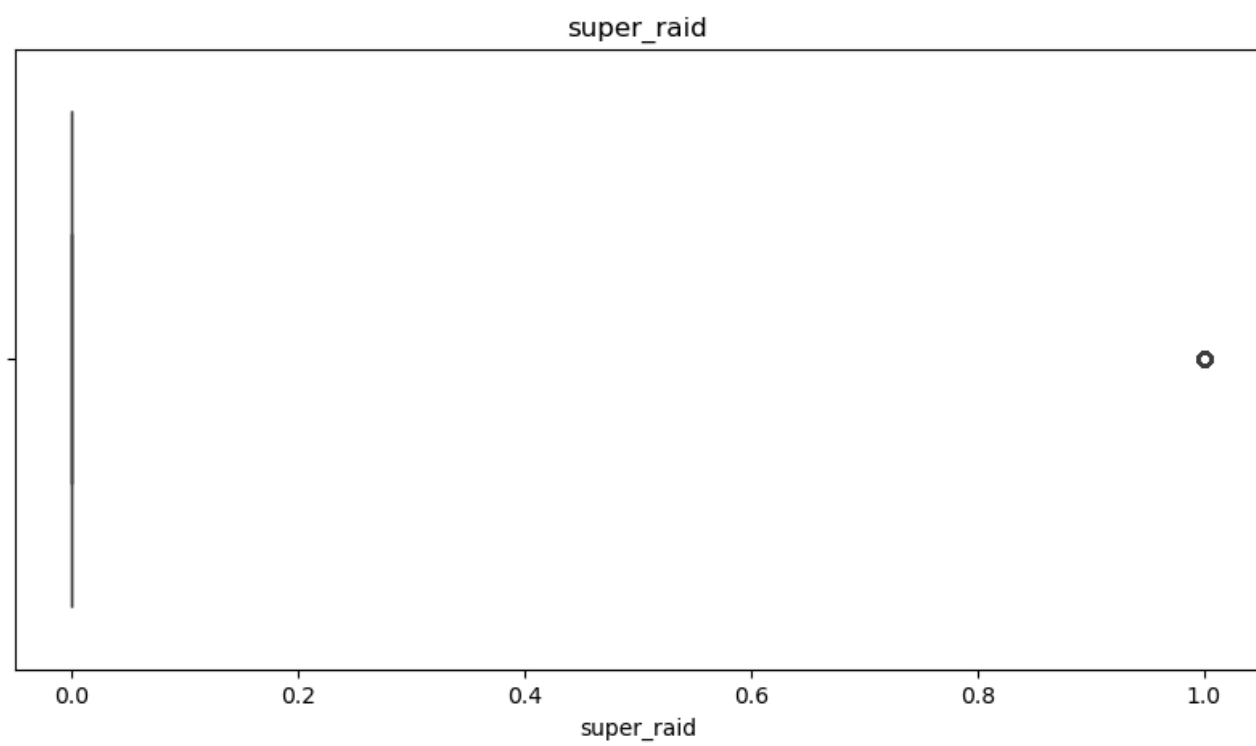


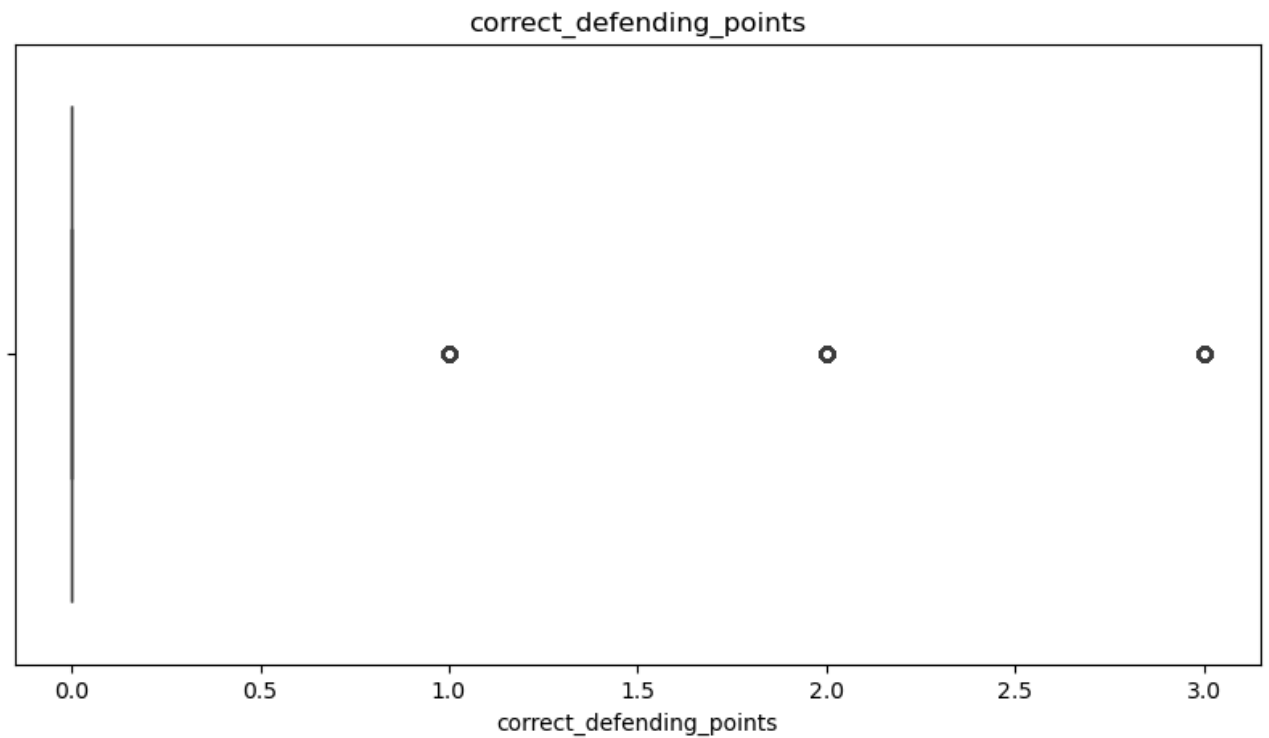












```
In [12... # import seaborn as sns
# import matplotlib.pyplot as plt

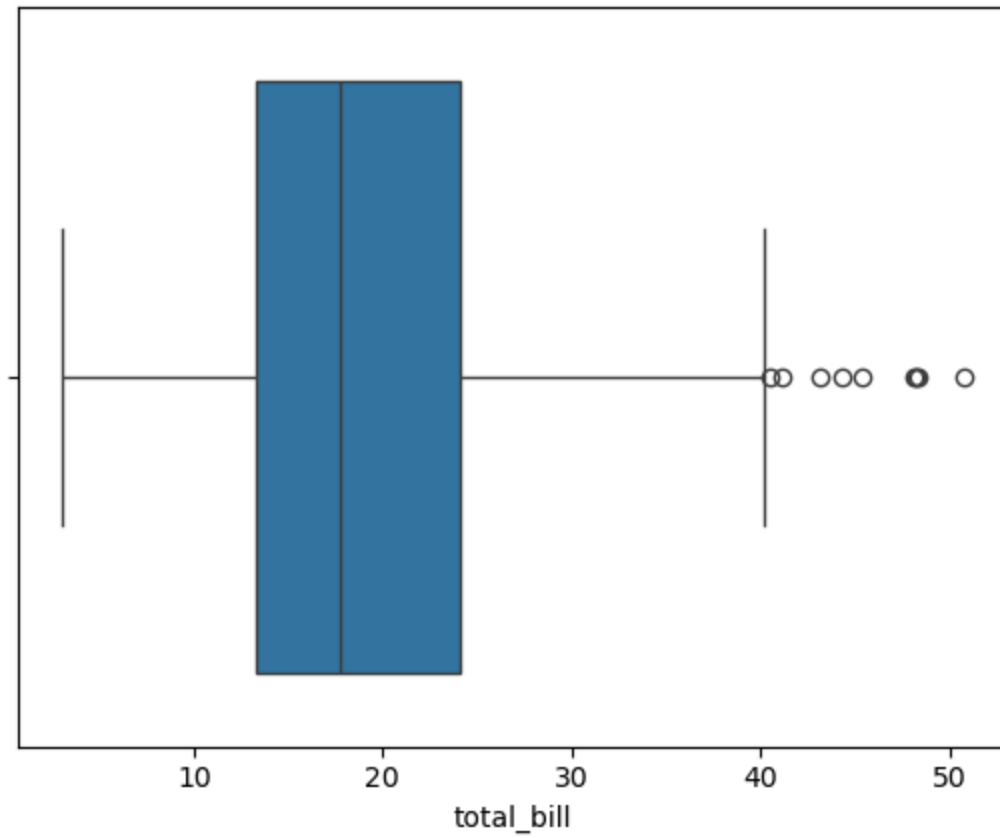
# df = sns.load_dataset("tips")

# print(df.head())
# print(df.describe())
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
In [12... # sns.boxplot(x=df["total_bill"])
# plt.show()
```



```
In [ ]: #IQR=Q3-Q1
        #lower bound=Q1-1.5*IQR
        #upper bound=Q3+1.5*IQR
        #task--->numerical column
```